

Rebecca has \$1.90 worth of nickels and dimes. She has 2 more nickels than dimes.

1. Define the variables to write the system of equations.

Let _____ = _____

Let _____ = _____

2. Write a system of equations to determine the number of nickels and dimes Rebecca has.

3. Solve the system using the method of your choice.

4. Rebecca has _____ nickels and _____ dimes.

Sampson and Riki went to the movies to watch 'The Batman' that was just released. Sampson bought 2 bags of popcorn and 5 pretzels, costing a total of \$37.25. Riki bought 8 bags of popcorn and 9 pretzels, costing a total of \$102.25. Determine the price of each bag of popcorn and the price of each pretzel.

5. Sampson created the equation $2b + 5p = 37.25$. Create a second equation that could be used to find the price of each bag of popcorn and the price of each pretzel.

6. Solve the system using the method of your choice.

7. The price of each bag of popcorn is _____. The price of each pretzel is _____.

A new Marvel movie was released at the Imax theater. The Friday 8 P.M. showing sold a total of 150 tickets. A child ticket costs \$7.75 and an adult ticket costs \$10.25. The show generated a total of \$1,470.

8. Write a system of equations that could be used to find the number of child tickets that were sold and the number of adult tickets that were sold.

9. Solve the system of equations you wrote.

Dale wants to build a new deck for his house. The nails come in large boxes and small boxes. The large boxes each have 25 nails, and the small boxes hold 20 nails. Dale bought a total of 295 nails and a total of 13 boxes.

10. Write a system of equations to find how many boxes of small and large boxes of nails Dale bought.

11. Solve the system of equations you wrote.